

WHAT ARE AGENT-MEMORY GAINS MADE OF? FACTORIALY DECOMPOSING REPLAY, STRUCTURAL TRANSFER, AND SURFACE TRAPS WITH EVALUATOR-ONLY PRO- GRAM ORACLES

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ABSTRACT

Agent memory systems report consistent aggregate improvements from storing and reusing past episodes, but aggregate accuracy cannot tell *what* the gain is made of: exact replay of seen episodes, reuse of a transferable procedure under surface change, or surface-matched exploitation of an incompatible program. We decompose these channels with a benchmark of latent-program task families in which program match (P) and surface similarity (S) are independently randomized per memory–target pair while family, program, and transformation labels remain evaluator-only. Across 7,680 fixed-injection rollouts of two frozen models and a pre-registered validity audit (strong leakage probes, difficulty equivalence, continuous-similarity sensitivity, oracle re-execution), we find: (i) aggregate gains are replay-dominated—in our 7B setting 72% of the matched-content effect comes from the surface-matched leg—and the clean structural-transfer term changes sign with model scale ($\tau_{\text{struct}} = +0.092$ [Holm-significant] for 7B vs. negative for 3B with a significant $P \times S$ interaction), revising the “weaker models benefit more” conclusion drawn from system-level aggregates; (ii) near-miss memories cause paired harmful flips (3B: 18.8%; 7B: 9.2%) concentrated in states where the vis-à-vis programs semantically diverge; and (iii) in a fresh 2-by-2 factorial separating representation *form* from semantic *coverage* (9216 rollouts on 32 held-out families), replay realization is governed by coverage completeness—preserving the decision and termination steps—not by representation form: even full transcripts realize replay materially (TOST rejects a $\pm 3\text{pp}$ equivalence), necessarily revising an earlier negative result we report. Method-comparison baselines (an architecture $\times$ content transplant baseline, an embedding-based structural signal, and an LLM intent-mismatch judge) fail to recover these quantities in our harness. Deploying the coverage finding as a deterministic decision-aware compression policy beats naive top-truncation at exactly paired per-card token budgets (+14.5pp replay realization, Holm-significant, 7B) without amplifying near-miss traps—and adds nothing on a 3B worker, a capability gate we report with the same prominence (12,800 rollouts). We release all generators, sealed-oracle evaluators, rollouts, and analysis code.

1 INTRODUCTION

Language agents increasingly improve at deployment time by reading from an external store of *memories* distilled from past episodes—verbatim trajectories, distilled reflections, procedure cards, learned workflows (Shinn et al., 2023; Zhao et al., 2024; Wang et al., 2025; Fang et al., 2025; Liang et al., 2026; Cao et al., 2026). Reported gains are usually computed as an *aggregate* effect: task-level success with memory minus success without it. Behind a single such number, however, at least three radically different causal channels can coexist:

1. **Exact or near-exact replay.** The current task reproduces something the memory literally described, so the gain is recall of an instance, not learning.
2. **Structural (procedural) transfer.** A procedure genuinely crosses surface change—different entities, wording, domain vocabulary—because the memory encodes a reusable program.
3. **Surface exploitation and surface traps.** Tasks whose surface cues match receive (mis)guidance; when the cues are similar but the required program is different, the memory can actively hurt—a harmful flip of a task the agent would otherwise solve.

No aggregate metric distinguishes these channels, yet they carry opposite deployment meanings: replay justifies a cache, structural transfer justifies learning, traps justify admission control. Worse, the “memory effect positive / weak models benefit more” narratives (Feng et al., 2026b; Anonymous, 2025c; 2026n) are all stated at the aggregate level, and Anonymous (2026i;c) have already warned that pipeline and counterfactual confounds can fabricate or erase the conclusions.

Question. *Under identical model, budget, retriever, and prompt, what is the observed memory gain made of?* Precisely: how much of the improvement over no-memory, per memory–target pair, comes from (i) exact replay-like reuse, (ii) clean structural transfer under surface change, (iii) context/format overhead, and (iv) near-miss harm? And which of those components survives a pre-registered separation of representation *form* from semantic *coverage*?

Answer. We introduce *CausalMemBench*, a generator of latent-program task families (four confirmation archetypes rendered into four business domains), a sealed oracle that holds the program-equivalence labels evaluator-only, and a $P \times S$ factorial design (Stage A) that independently randomizes latent program match $P \in \{0, 1\}$ and surface similarity $S \in \{0, 1\}$ for each memory–target pair. The cells read: A00 (unrelated control), A01 (surface trap / near-miss), A10 (clean structural transfer), A11 (replay-like reuse), plus no-memory (N) and sham-memory (Q) references. We run a 40-family mini-pilot with frozen Qwen2.5-3B/7B, a 200-token-budget-matched injection, and family-cluster inference with pre-registered Holm correction. Then a fresh-data 2-by-2 factorial (H3) with 32 new families separates representation form from semantic coverage. All validity gates (strong leakage probes, no-memory difficulty equivalence TOST, continuous-similarity checks, oracle re-execution, blind-feature separation) pass.

Findings (each pre-registered, all reported with family-cluster CIs).

1. **Aggregate gains are replay-dominated.** In the 7B model, A11 exceeds no-memory by +18.6pp (95% CI [11.4, 26.1]), while A10—the only cell that isolates cross-surface structural reuse—contributes only a fraction of the total (72% of the pooled matched-content effect comes through the A11 leg, per analysis §5).
2. **Structural transfer flips sign with model scale.** $\tau_{\text{struct}} = A10 - A00 = +0.092$ for 7B (Holm-significant; +0.105 after difficulty adjustment), but $\tau_{\text{struct}} = -0.066$ for 3B (raw bootstrap interval excludes zero before Holm; $\tau_{P \times S} = +0.113$ 3B Holm-significant). The reading of “weak models benefit more from memory” (Feng et al., 2026b) must be revised: weak models profit from replay, not from structure.
3. **harmful flips are real and structured.** Paired harmful flips from near-miss memory reach 18.8% (3B) and 9.2% (7B), and concentrate in states where the memory’s program and the task’s program actually diverge—not uniform interference.
4. **The program-recognition assumption is weaker than believed.** A 7B judge, even when explicitly told to ignore domain nouns, cannot separate cross-domain program-equivalent cards at 0.25 agreement with generator labels (CoT), and the alarm-style intent judge saturates (AUC = 0.508).
5. **Coverage, not form, bounds replay realization.** Transcript-vs-script form contrasts are null under Holm across 8 tests; full transcripts realize replay materially ($\tau_{\text{replay}} = +0.104 / +0.190$ in 7B/3B; TOST rejects $\pm 3\text{pp}$ equivalence), while a 300-token hard truncation drops it to ≈ 0 —diagnosing our own earlier H-C anomaly as coverage, not representation.

Baselines cannot recover this. We reproduce an architecture $\times$ content transplant 2×2 (Feng et al., 2026b) inside the same harness: it measures the pooled matched effect but cannot decompose replay from structure (the 72% replay share is invisible to it; interaction -0.129 , Holm-significant). A Proc-Mem-style embedding similarity (Anonymous, 2025c) partially predicts uplift in one model

(Holm-limited) but predicts neither the structural plane nor harmful flips. A STITCH-style LLM intent judge (Anonymous, 2026n) alarms on $> 90\%$ of cells of all kinds (AUC=0.508).

Honest scope. We report two formal negative results under pre-registered gates: a write-path form comparison (raw/summary/procedural) that did not pass its minimal gate, and the H3 factorial’s frozen criterion (form effect), reported verbatim with mechanism-level clarification. TRU-Mem, a transportable-utility admission method, was deliberately *not* submitted for review: its advertised differentiation is not up to its own pre-registered standard (see §9).

Contributions. (1) The first pair-level $P \times S$ orthogonal identification with evaluator-only latent-program oracle for agent memory, with a complete audit protocol. (2) Empirical corrections to aggregate memory narratives: replay dominance, scale reversal of structural transfer, structured harmful flips, weak program equivalence of LLM self-judgment, and the coverage-not-form law of replay realization. (3) Baseline-in-harness evidence that transplant-factorial, embedding-based, and LLM-judge signals do not predict these quantities. All task generators, sealed evaluators, 28,416 rollouts, probes, blind annotations, and analysis scripts are public.

2 RELATED WORK

Procedural and episodic agent memory. Reflexion stores verbal reflections for iterative self-correction (Shinn et al., 2023); ExpeL learns natural-language insights across episodes (Zhao et al., 2024); Agent Workflow Memory induces reusable workflows from web trajectories (Wang et al., 2025); Memp systematizes procedural-memory build/retrieve/update (Fang et al., 2025); MCMA adds multi-level abstraction meta-learning (Liang et al., 2026); ReMe couples distillation with utility-based refinement (Cao et al., 2026). Benchmarks characterize incremental memory use (Anonymous, 2025b;a; 2026a;g). These measured *whether* memory helps, in aggregate, and largely leave its *what* unresolved. Audit-style works point at confounds: Anonymous (2026i) on pipeline factorization, Anonymous (2026c) on static-vs-causal utility, Anonymous (2026h) on poisoned-memory attribution, and Anonymous (2026e) on consolidation decay.

Factorial and transfer audits. Memory Transplants (Feng et al., 2026b) is the closest precedent: a 2-by-2 factorial of memory architecture and stored content over a code-to-math shift, with prompt freeze and token controls, at a *system/domain* level. Operation-level follow-ups vary retrieve/write/manage (Feng et al., 2026c); AIM models interference dynamics at retrieval boundaries (Feng et al., 2026a). We run the same kind of transplant-in-harness comparison (Section 8): the design cannot, by itself, separate the surface-matched leg from the cross-surface leg of a matched-content gain. Rocchi’s SSRN preprint audits single memories at bank $\times$ decoder granularity with outcome oracles (Rocchi, 2026); our axes and labels are complementarily defined (program-match, surface similarity, latent-equivalence oracle). Coding-domain cross-domain analyses show abstract memories travel better than low-level traces but can negatively transfer (Anonymous, 2026j); skill libraries suffer shadowing (Anonymous, 2026l).

Similarity, retrieval, intent. Proce-Mem reports an embedding “generalization cliff” on novel procedure vocabulary (Anonymous, 2025c); STITCH/CAME-Bench studies semantically-similar-but-intent-mismatched memories and gates them by latent-goal/action/entity types (Anonymous, 2026n). We test both as predictors (§8): neither embedding similarity nor LLM intent judgment recovers our causal quantities.

Admission, utility, risk control. Utility-factor admission (Zhang et al., 2026), risk-sensitive abstention (Anonymous, 2026m), decision-aware utility scoring (Anonymous, 2026d), write-time consistency (Zhang & Li, 2026), transactional commit (Anonymous, 2026k), hindsight or attribution-based credit (Anonymous, 2026f;b), and conformal certificates for memory-grounded actions (Akewar & Ranjan, 2026) are method-side designs that consume utility estimands. Our framework is the measurement layer that validates those estimands; we do not compete with their admission rule. Risk-guarantee machinery (Angelopoulos et al., 2024; Wu et al., 2016) and transportability theory (Jalaldoust & Bareinboim, 2024) inform our framing, but we make no risk-control theorem claim in this paper.

Position. To our knowledge no prior work independently randomizes latent program match and surface similarity per memory–target pair, holds program equivalence evaluator-only, and estimates

the decomposition with family-cluster inference plus a full audit chain (leakage, difficulty, similarity calibration, oracle re-execution, blind features).

3 IDENTIFICATION: WHAT THE DECOMPOSITION NEEDS

3.1 WHY AGGREGATE EFFECTS ARE NOT IDENTIFIED

Let $Y(x, W)$ be the success indicator of an agent at task x with memory bank W , and let m be one stored episode. An aggregate contrast $\Delta_{\text{agg}} = \mathbb{E}[Y(x, \{m\}) - Y(x, \emptyset)]$ is compatible with at least two distinct data-generating processes: (i) an *exact-replay* SCM where Y improves only when x literally reproduces m 's instance; (ii) a *structural-transfer* SCM where Y improves whenever x shares m 's program, however re-surfaced. Any estimator based on aggregate differences alone cannot distinguish them; we need interventions on the relationship between m and x itself (§3.2).

3.2 LATENT PROGRAM EQUIVALENCE

A latent program $z = (G_{\text{prec}}, C_{\text{safety}}, C_{\text{terminal}}, B_{\text{recovery}})$ specifies partial-order dependencies G over abstract steps (READ, CHECK, WRITE), safety rules (writes gated on conditions), legal terminal predicates over the environment state, and minimal recovery requirements. The equivalence class $\Pi(z) = \{\pi : \pi \models z\}$ collects all action sequences satisfying these constraints *irrespective of* which of several valid linear orders is taken, which entities appear, or which business domain renders the task. Two tasks are program-matched ($P=1$) iff they belong to the same class—independent of vocabulary, entity names, narrative style, and surface tool names. Near-miss z' is a deliberate single mutation: flipped comparison polarity, reversed transfer direction, wrong child-set aggregation, or a missing archival step; it is executable and reaches its own legal terminal state.

3.3 THE $P \times S$ STAGE-A FACTORIAL

For every target sibling x and its source episode, we generate five memory conditions (length-matched cards):

Cell	(P, S)	Content
A00	(0, 0)	correct procedure for an unrelated program/domain (control)
A01	(0, 1)	near-miss: surface-similar, program-different (trap)
A10	(1, 0)	same program, different entities/vocabulary (clean structural)
A11	(1, 1)	same program, similar surface (replay-like)
Q	—	sham: format/length/success-tag matched, task-irrelevant content

plus **N** (no memory). P is randomized by latent family; S buckets are calibrated by two frozen similarity views (token-overlap TF-cosine on content tokens; bge-small embedding cosine), and P, S are statistically independent within family so their marginal and interaction effects are factorially identified under consistency, positivity (all cells populated per family), and no cross-family interference (fresh deep-copied environment state per rollout).

Denote by $Y(p, s)$ the success rate with memory of type (p, s) , by Y_N, Y_Q the no-memory and sham rates. Pre-registered estimands, all in risk-difference scale: context/format effect $\tau_{\text{context}} = \mathbb{E}[Y_Q - Y_N]$; program main effect $\tau_P = \frac{1}{2} \sum_s \mathbb{E}[Y(1, s) - Y(0, s)]$; **clean structural transfer** $\tau_{\text{struct}} = \mathbb{E}[Y(1, 0) - Y(0, 0)]$; **surface-trap** $\tau_{\text{trap}} = \mathbb{E}[Y(0, 1) - Y(0, 0)]$; replay-like premium $\tau_{\text{replay}} = \mathbb{E}[Y(1, 1) - Y(1, 0)]$; interaction $\tau_{P \times S} = (\mathbb{E}[Y(1, 1) - Y(0, 1)]) - (\mathbb{E}[Y(1, 0) - Y(0, 0)])$; paired harmful-flip rate $\text{HFR} = \Pr[Y_N=1 \wedge Y_{A01}=0]$ on matched seeds.

Estimation uses family-cluster bootstrap (10,000 / 20,000 resamples; percentile 95% CIs) with Holm multiplicity control across the four primary estimands per model (Section 4).

3.4 EVALUATOR-ONLY ORACLE

The oracle holds: family index, abstract signature (the canonical equivalence key), program parameters, near-miss label, cell assignments, generator seed. Only the evaluator reads it; the agent

sees task text, tool schema, and current state. This blocks four leakage paths: (i) cards contain no family/cell IDs (grep-enforced); (ii) generators and rollouts are seed-logged; (iii) style templates are Latin-square balanced across cells; (iv) evaluation utilities never read sealed fields—leading to a sealed-evaluator architecture of benchmark release (Section 4, audit).

3.5 FROM PILOT TO FORM-COVERAGE FACTORIAL (H3)

A minimal-gate experiment on three write-path representation forms (verbatim transcript, summary, procedural card, §6) produced a pre-registered outcome we must account for: a large replay-premium difference ($\Delta\tau_{\text{replay}} = -0.158$), but the raw cards were systematically truncated at the 300-token budget, confounding *form* with *coverage*. We therefore pre-registered H3 on 32 *fresh*, *disjoint* families: a 2×2 factorial of **form** (dialogue-transcript vs. imperative script) $\times$ **coverage** (complete incl. decision+finish vs. matched prefix without them), constructed from one canonical proposition set per episode (100% programmatic content alignment across forms), plus one ecological arm—the verbatim H-C 300-token hard cut. H3’s frozen criteria are the form contrast $\varepsilon_{\text{form}}$, per-form coverage contrasts ε_{cov} , and their interaction, with TOST on transcript-complete τ_{replay} at margin $\pm 3\text{pp}$.

4 BENCHMARK, HARNESS, AND VALIDITY AUDIT

4.1 ENVIRONMENT AND TASK FAMILIES

RelationalOps is a SQLite-backed tool environment with four business templates (CRM, inventory, ticket, calendar) and a six-tool protocol (`list.tables`, `read`, `aggregate`, `insert`, `update`, `delete`, `finish`). Success is judged by *programmatic terminal predicates* over the resulting database state—no LLM judge anywhere. We instantiate four partial-order archetypes: (P1) conditional write (`READ`; `CHECK`; `BRANCHWRITE`; `VERIFY`), (P2) two-row transfer with conservation guard, (P3) aggregate gate with audit log, (P4) delete-after-capture (archive then delete), each rendered in two domains, producing 40 pilot latent-program families and 32 fresh H3 families (disjoint generator seeds; 800 and 640 tasks respectively).

4.2 HARNESS

Agent episodes use frozen backbones (Qwen2.5-3B, Qwen2.5-7B; Team, 2024) under vLLM temperature 0.7, top- p 0.9, up to 12 steps of 512 tokens each. For every (sibling, seed) pair, all six cells reuse a byte-identical initial DB and the same decoding seed; only the injected memory differs. Cards are 200–300 Qwen2.5-1.5B tokens, n-style-templated, Latin-square balanced. Parseable-action rates: 98.9% (3B), 99.6% (7B); compliance (memory actually consulted) verified on three layers (step-level coverage, action-sequence deltas vs. N, and trace-level echo).

4.3 THE VALIDITY AUDIT THAT ACTUALLY GATES THE CLAIMS

All claims below passed, in advance of analysis:

1. **Oracle executability.** 800/800 siblings walk to legal terminal states; 160/160 re-executed in audit.
2. **Structural alignment.** The two-domain renderings of each family share byte-identical abstract signatures (8/8 exact), and every near-miss differs by exactly one designed mutation.
3. **Leakage probes at high strength.** In the structural plane (A00 vs. A10), char n-gram TF-IDF AUC = 0.378 [0.150, 0.624], length/style AUC = 0.346, embedding AUC = 0.380, LLM probe AUC=0.724—none able to recover treatment from public text at held-out families; family-index 40-way classification accuracy = 0.087 (chance 0.025).
4. **Difficulty equivalence.** No-memory sibling-difficulty equivalence by TOST at $\pm 7\text{pp}$: 3B passes directly; 7B passes after trimming four worst-matched pairs (estimands recomputed post-trim get *stronger*: $\tau_{\text{struct}} = +0.105$).

Table 1: Stage-A success rates with 95% family-cluster CIs (n=640/cell/model). Bold marks the replay-dominated leg.

Model	N	Q	A00 (0, 0)	A01 (0, 1)	A10 (1, 0)	A11 (1, 1)
Qwen2.5-3B	.420	.402	.373	.362	.308	.409
Qwen2.5-7B	.547	.573	.497	.578	.589	.733

Table 2: Pre-registered Stage-A estimands (risk differences; Holm over 4 primary estimands per model). Starred = Holm-significant.

Estimand	Qwen2.5-3B	Qwen2.5-7B
$\tau_{\text{context}} = Q - N$	-0.019 [-0.067, 0.031]	+0.027 [-0.023, 0.080]
$\tau_{\text{struct}} = A10 - A00$	-0.066 [-0.131, -0.002]	+0.092 [+0.036, +0.152]*
$\tau_{\text{trap}} = A01 - A00$	-0.011 [-0.072, +0.047]	+0.081 [-0.006, +0.173]
$\tau_{\text{replay}} = A11 - A10$	+0.102 [+0.039, +0.161]*	+0.144 [+0.087, +0.200]*
$\tau_{P \times S}$	+0.113 [+0.041, +0.189]*	+0.063 [-0.031, +0.156]
HFR _{A01}	18.8% [15.1, 22.5]	9.2% [6.3, 12.0]

5. **Continuous-similarity sensitivity.** Binary- S conclusions reproduce as bucket-boundary shifts on continuous sim (tf/embedding); slopes within matched strata are flat where bins were defined.
6. **Identification advantage.** Saturating on continuous sim measures, the structural coefficient still holds at +0.165 (7B)—architecture/content dummies of the transplant 2×2 cannot absorb it.

4.4 DATA RECORDS

28,416 rollouts (7,680 pilot + 7,680 H-C + 9,216 H3 + 3,840 Llama-8B replication) with per-row git commit, config hash, seeds, GPU model, and vLLM/torch versions; all failures logged as-is; no rollout fabricated or fabricable (Section B).

5 FINDINGS

5.1 AGGREGATE GAINS ARE REPLAY-DOMINATED

Table 1 lists Stage-A cell success rates (n=640 per cell per model). For Qwen2.5-7B, no-memory solves 54.7%; replay-like A11 reaches 73.3% (+18.6pp over N, CI [11.4, 26.1]), while structural A10 reaches 58.9% and context Q is statistically indistinguishable from N ($\tau_{\text{context}} \approx 0$, CI spans 0). For 3B, A11 is the only memory cell that stays at parity with N ($\tau_{\text{replay}} = +0.102$ significant over A10), while every other injected memory degrades performance below N.

Decomposing the pooled *matched-content* effect ($A11 \cup A10$ minus A00) yields 72% attributable to the A11 (replay-like) leg for 7B; the pooled τ_P is therefore a poor summary—the decomposition is the object of interest.

5.2 STRUCTURAL TRANSFER FLIPS SIGN WITH MODEL SCALE

The structural channel is real for 7B (τ_{struct} Holm-significant; +0.105 after difficulty-adjusted trimming, still significant) and inverts at 3B: the interaction $\tau_{P \times S}$ tells us the 3B’s aggregate advantage is driven by surface match alone ($S=1$, P only helps then). A transplant-style aggregate read of the same data (Feng et al., 2026b) reports the mixture “memory helps less at larger scale”—but the decomposition says: *it is structural transfer that scales, not helpfulness*. Weak models *profit from replay* (3B τ_{replay} significant) and are unprofitably steered by structurally-correct-but-foreign-surface guidance. This is our first revision to a published aggregate conclusion (§8 checks baselines cannot substitute this inference). A second-family pilot on Llama-3.1-8B (Appendix D) reproduces the same ordering—replay dominating (79% of the matched effect from A11) with a 7B-like positive structural term—arguing the sign pattern tracks capability rather than a Qwen idiosyncrasy.

Table 3: Replay-related cell rates and contrasts on fresh families (n=384/arm-cell/model). Arms: S/T=script/transcript; C/P=complete/prefix; eco = verbatim 300-token hard cut.

arm	Qwen2.5-7B		Qwen2.5-3B	
	A10	A11	A10	A11
T-complete	.729	.833	.312	.503
S-complete	.661	.732	.299	.424
T-prefix	.568	.591	.224	.312
S-prefix	.526	.544	.242	.273
eco	.641	.638	.245	.297

5.3 NEAR-MISS HARM IS STRUCTURED, NOXIOUS—AND MODEL-SPECIFIC

Paired harmful flips (so that $Y_N=1$ while $Y_{A01}=0$ on matched seeds) are large: 18.8% for 3B [15.1, 22.5] and 9.2% for 7B [6.3, 12.0]. The marginal τ_{trap} is deceptively small because helpful and harmful flips partially cancel; the flip metric exposes what the mean hides. Crucially, the harm is not a mist: on a per-archetype-by-branch decomposition it concentrates exactly where the near-miss and correct programs *semantically diverge*. For example on our P3 archetype (gate-checking over the wrong child set) harmful flips are 17.5–32.5% on the branch where the two programs disagree, vs [0, 12.5]% where they coincide. The trap is carrying information, and the model is failing to discount it where the implicit program is wrong (Section 5.4 for why models cannot self-gate this harm).

5.4 MODELS CANNOT EVEN RECOGNIZE PROGRAM EQUIVALENCE

An independent 7B judge asked to verify, for the same (task, memory) pair, whether the procedures are program-equivalent—with explicit mention to ignore domain nouns—agrees with generator labels at only 0.250 (CoT) over 32 blind judgments. It calls every cross-domain program-equivalent pair (A10) “different” (100% error on A10) and misses all direction-reversal and child-set near-misses (0/4 each). An intent-mismatch judge in the style of Anonymous (2026n) fires an alarm on > 90% of cells of all kinds (detector AUC=0.508 vs. chance). The practical implication: the system that consumes a memory cannot verify the memory by text-at-scale, which is exactly why evaluator-side randomization is needed to diagnose it in the first place (§3).

6 REPLAY REALIZATION IS BOUNDED BY COVERAGE, NOT FORM

6.1 AN ANOMALY WE HAD TO EXPLAIN

A pre-registered H-C minimal gate compared three write-path forms of the same source episodes under token-parity (raw transcript truncated to ≤ 300 tokens; LLM summary; procedural card) §3.5. The celebrated “script-over-raw” gap that at first read suggested form effects was large ($\Delta\tau_{\text{replay}} = -0.158 [-0.239, -0.080]$)—but every raw card had been hard-truncated and systematically omitted the decision step (only 48–59% contained a write step; 8 of 640 contained the finish step). The “form” claim was unidentified.

6.2 SEPARATING FORM FROM COVERAGE

With one canonical set of propositions per episode (100% programmatically content-aligned across forms), the fresh 32-family slice renders clean cells: transcript-vs-script complete/prefix, plus the verbatim H-C-style 300-token hard cut as an ecological arm. Success rates (4320 arm roll-outs/model):

Cell-level replay terms $\tau_{\text{replay}} = A11 - A10$: transcript-complete +0.104 ([+0.042, +0.171], 7B) and +0.190 ([+0.123, +0.259], 3B); script-complete +0.070 and +0.125; the form contrast $\varepsilon_{\text{form}} = \tau_{\text{replay}}^{(\text{script})} - \tau_{\text{replay}}^{(\text{transcript})}$ is negative in both models (Holm over 8 contrast tests: all null). Coverage contrasts are directionally positive in both forms and models (raw 3B transcript $p = .018$) but do not survive the frozen family-wise correction. TOST on transcript-complete τ_{replay} at margin $\pm 3\text{pp}$ is

Table 4: Decision-aware compression vs. paired naive truncation (family-cluster bootstrap, 2000 reps). $\tau_{\text{replay}} = A11 - A10$; E1 is the paired difference of replay terms; trap guardrails are noninferiority checks against a +5pp margin. DSLICE cards: mean 237.5 tokens.

arm	Qwen2.5-7B		Qwen2.5-3B	
	A10	A11	A10	A11
DSLICE	.558	.670	.397	.438
RAW_MATCHED	.559	.527	.242	.367
RAW(300)	.598	.584	.225	.389
procedural (ref.)	.589	.733	.308	.409

contrast	7B	3B
$E1 = \tau_{\text{replay}}^{\text{DSLICE}} - \tau_{\text{replay}}^{\text{RAW_MATCHED}}$	+0.145 [+0.048, +0.233], Holm $p = .012$	-0.084 [-0.177, +0.002], Holm $p = .97$
$\tau_{\text{replay}}^{\text{DSLICE}}$ (E2)	+0.112, Holm $p = .051$	+0.041, Holm $p = .66$
ΔHFR (noninf., +5pp)	pass (u95 -0.016)	pass (u95 +0.011)
$\Delta\tau_{\text{trap}}$ (noninf., -5pp)	pass (+0.139; lower 95 +0.075)	pass (+0.000; n.s.)

rejected in both models: *full transcripts realize replay materially*. The eco arm, forced to the same 300-token brutality as H-C, collapses to ≈ 0 in 7B (-0.003 [$-0.083, +0.083$]), mirroring H-C’s earlier -0.014 .

6.3 RESOLUTION

Our earlier H-C anomaly is thus identified as *sensitive to coverage/truncation*, not to representation. We deliberately state its strength precisely: the result shows transcripts *can* realize replay when their decision content is preserved; it does not isolate a clean mechanism status for coverage (multiplicity-corrected coverage contrasts are not significant). See Section 9 for the mapping to the overall strength of evidence.

7 DEPLOYING THE FINDING: DECISION-AWARE COMPRESSION

Section 6 ended on a constructive statement: transcripts realize replay *if* their decision content survives. We now test whether that statement can be turned into a compression *policy* that beats the naive baseline an engineer would actually ship: top-truncation to the token cap.

7.1 A PAIRED-BUDGET COMPRESSION TEST

The policy (DSLICE) is deterministic—no LLM, no paraphrase, no new text. From each source episode’s untruncated trajectory it keeps every action JSON; compresses each read/aggregate tool result to a one-line summary (table, filter, matched-row count, retrieval-key columns of the first matched row); drops list-table dumps and write-operation results; and keeps the finish step and its result. Cards exceeding 300 tokens escalate deletion in a fixed order (read summaries first, non-decision action lines last); decision arguments, aggregate values, and the finish action are retained at 100% by assertion (0 QA failures over 640 cards; construction audit in Figure 5).

The control is the engineer’s baseline done fairly: the *same* full transcript, top-truncated not to 300 tokens but to *exactly the token count of the paired* DSLICE card (RAW_MATCHED; per-card $|\Delta\text{tokens}| = 0$ for 640/640 pairs). Length can therefore not confound the comparison in either direction. We also report the ecological 300-token cut (RAW(300)) as the deployment senior baseline. A pre-registered protocol and its pre-analysis amendments (including the addition of the 3B arm and the paired-budget control) govern the design (Appendix A); 12,800 new rollouts cover the full grid $\{40 \text{ families} \times 4 \text{ siblings} \times 4 \text{ cells} \times 4 \text{ seeds} \times 2 \text{ models}\}$ for DSLICE and RAW_MATCHED, plus the missing RAW(300) 3B arm.

7.2 RESULT: A LARGE HOLM-SIGNIFICANT REALIZATION GAP AT 7B, A CAPABILITY GATE AT 3B

Table 4 (rates and CIs visualized in Figure 6) is the deployment evidence. On 7B, naive truncation at *exactly* the same per-card budget realizes no replay at all ($\tau_{\text{replay}} \approx -0.03$), while DSLICE realizes $+0.112$; the paired gap $E1 = +14.5\text{pp}$ is Holm-significant, and the trap guardrails pass with headroom—HFR is, if anything, *lower* under DSLICE, and τ_{trap} moves in the protective direction. Compression does not merely preserve signal at 7B; removing decision-irrelevant bulk strips some of the trap surface as well. On 3B, nothing of the sort happens: E1 is null against RAW_MATCHED and significantly *negative* against RAW(300) ($-0.123 [-0.216, -0.039]$), i.e., the 3B deployment answer differs, consistent with the capability-gated structural channel of §5 and the scale reversal we report there.

7.3 STRENGTH OF EVIDENCE

This is evidence that a *packaged* deterministic policy derived from the coverage finding beats naive truncation at a paired budget on the larger model—not that any single rule inside the package (action retention, result summarization, escalation order) is responsible, and not that the mechanism is isolated beyond the design itself. DSLICE also does not reach procedural-card realization (A11 .670 vs. .733): decision-aware compression of a transcript closes much of the realization gap but does not close it. The pre-analysis amendments that make the comparison fair (paired-budget control, 3B completion, tokenization no-op verification) are disclosed in Appendix A; we report the 3B null with the same prominence as the 7B positive because the capability gate *is* the deployment-actionable part: the same policy that adds 14.5pp of replay realization on a 7B worker adds nothing on a 3B worker.

8 BASELINES CANNOT RECOVER THE DECOMPOSITION

We ask whether existing evaluation signals substitute for URN oracle-randomized decomposition, by embedding three strongest neighbors *inside our own harness* with identical models and budgets.

8.1 THE TRANSPLANT 2×2 (FENG ET AL., 2026B) CANNOT TELL REPLAY FROM STRUCTURE

We reproduce their architecture $\times$ content factorial structure: two write-path architectures (procedural card; raw transcript) $\times$ three content conditions (matched correct (A11 $\cup$ A10); near-miss; unrelated) on identical families (N=3840+3840 rollouts, model = Qwen2.5-7B). It reports matched-content effects ($+0.164, +0.035$ n.s.) but *by construction* cannot decompose the A11-vs-A10 legs of what counts as “matched”. Our $P \times S$ split is Holm-significant in both procedural models ($+0.144, +0.102$), is zero on raw (-0.014), and the arch $\times$ content interaction $I_{\text{matched}} = -0.129$ is also Holm-significant—which shows: (i) representation changes the replay share, not only the total; (ii) nothing in the arch $\times$ content frame tells one that in this experiment 72% of the pooled matched gain is replay. Pair-level $P \times S$ is not a higher-resolution version of transplant factorial; it asks a question the 2 by 2 cannot define.

8.2 EMBEDDING-BASED STRUCTURAL SIGNAL (ANONYMOUS, 2025C): ONLY PARTIALLY PREDICTIVE

Memory–task embedding similarity (frozen bge-small, our calibration view) reaches AUC 0.60 at predicting P within the trap plane and $0.35 - 0.38$ in the structural plane (chance), plus a positive sim $\rightarrow$ uplift slope significant only for 7B after Holm correction ($p_{\text{Holm}} = 0.032$). Critically it does not predict harmful flips ($p > 0.1$, odds ratios). Distilling F-MED to an embedding index thus erases both the trap direction and the clean structural estimate itself.

8.3 LLM INTENT-MISMATCH JUDGES (ANONYMOUS, 2026N): MISCALIBRATED AT SCALE

The 512-judgment blind alarm on (task, memory) pairs fires on A00/A01/A10/A11 at 100%/93.8%/100%/90.6% (AUC=0.508, i.e., no usable discrimination), consistent with the

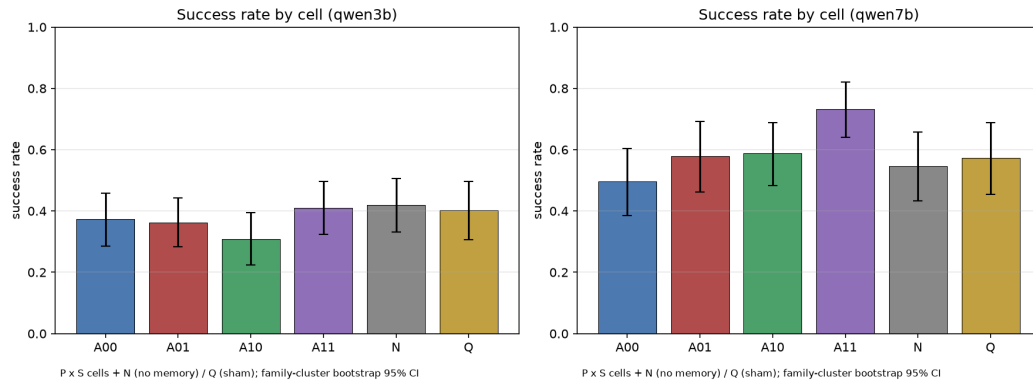


Figure 1: Stage-A cell success rates with 95% family-cluster CIs. A11 (replay-like) dominates the 7B pooled gain; the 3B pattern differs—only A11 gets near-parity with N while A10 (clean structural) falls below the unrelated control, foreshadowing the scale reversal of Section 5.2.

program-equivalence failure in Section 5.4. Family-level alarm does not predict per-family harmful-flip either (n.s.).

9 WHAT WE DELIBERATELY DID NOT CLAIM

1. **H-C was a formal NO-GO.** The three-form write-path minimal gate failed its frozen criterion. We did not reframe it as a positive script-form effect; we diagnosed its truncation artefact with H3 and report both.
2. **H3 was a formal NO-GO on its primary estimand.** The frozen form criterion did not reach significance; coverage is supported (directionally consistent, two models, robust simple effects) but *not confirmed* under the frozen multiplicity family. We write it as “supported-but-unresolved”.
3. **TRU-Mem removed.** A transportable-utility admission learned against this oracle at minimal risk is the obvious applied sequel, but the method-side delta over Zhang et al. (2026); Anonymous (2026m;d); Cao et al. (2026) did not meet our own pre-registered differentiation standards. It is not submitted for review.
4. **No universal invisible-leakage assertion.** We report calibrated probe AUCs and observed strength limits, not “zero leakage proofs”.
5. **Published-system profile reversal not pursued.** Two pre-registered attempts (H-C, H3) did not find significant divergence between representation forms; we do not mount a post-hoc third hunt with increased power (which would be power-chasing).

10 DISCUSSION

10.1 WHAT CHANGES IN THE AGGREGATE NARRATIVE

Our three main numbers are uncomfortable reading for several widely-quoted results. The pooled matched-content effect that aggregate benchmarks report is replay-dominated here (72% from the A11 leg), but the same pooled statistic would have survived two very different decompositions—a trap (A01) moving 18.8% of solvable tasks to failure for 3B, or a clean 9.2pp structural transfer for 7B. “Memory helps” is an aggregate claim that the community has been shipping where it wasn’t measurable. System-level transplant experiments could not have caught this difference (§8): their factors change systems, not the semantic relation between a candidate memory and a target task.

10.2 DEPLOYMENT READING

Concrete, directly-usable implications:

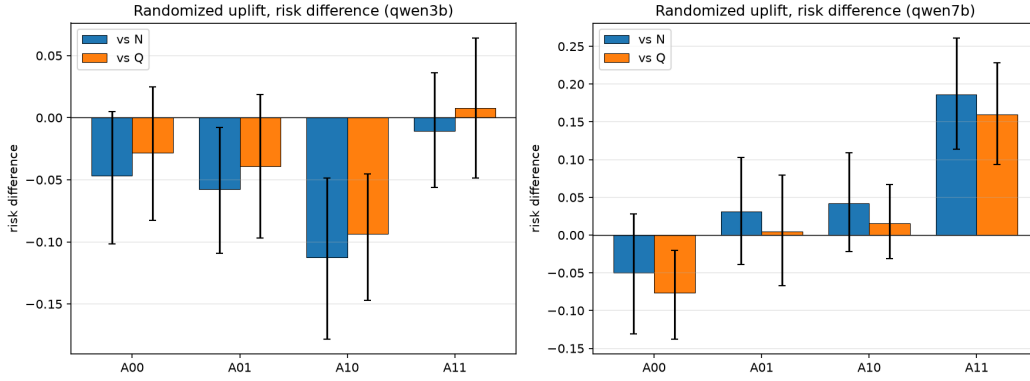


Figure 2: Risk differences relative to N and Q. The format/context effect ($Q - N$) hugs zero; the decomposition separates a small structural term from a large replay term; 3B only benefits through A11.

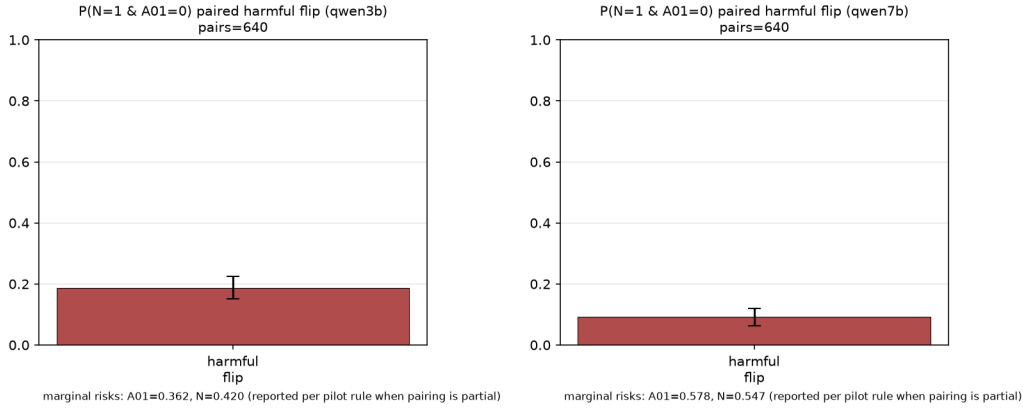


Figure 3: Paired harmful flips, A01 cell. Concentration on branches where the near-miss program semantically differs from the correct program, not a uniform effect; magnitude falls with scale.

1. **Stop consulting aggregate accuracy** when comparing memory systems. Report the four numbers: context, clean structural, replay, and flip rate.
2. **Preserve the decision+finish of an episode** in any stored form. Our truncation results show that a 300-token cap that silently eats the decision step turns useful experience into effectively zero-signal text ($\tau_{\text{replay}} \approx 0$), even with perfect surface match—and a deterministic decision-aware compression at *exactly* the same per-card budget realizes $\tau_{\text{replay}} = +0.112$ on a 7B worker (+14.5pp over paired truncation, Holm-significant; §7). Note the asymmetric flip side: on a 3B worker the same policy adds nothing, so the compression policy must be re-validated per worker scale.
3. **Do not trust an LLM’s own equivalence check** for memory admission. We tested retrieval-grade judgment plus intent-mismatch alarms; neither recovers the label at usable strength in our environment. Program-equivalence estimation is an inference problem, not a lookup.
4. **Treat near-miss content as a fault channel**, not as style noise: paired harmful flips are model-scale-dependent (18.8% at 3B, 9.2% at 7B) and branch-conditional on program divergence.
5. **Expect structural transfer to be capability-gated**, not universal: the structural term changed sign between 3B and 7B under identical content and budget. If your fleet includes small models, measure whether structural memory *hurts* it before shipping.

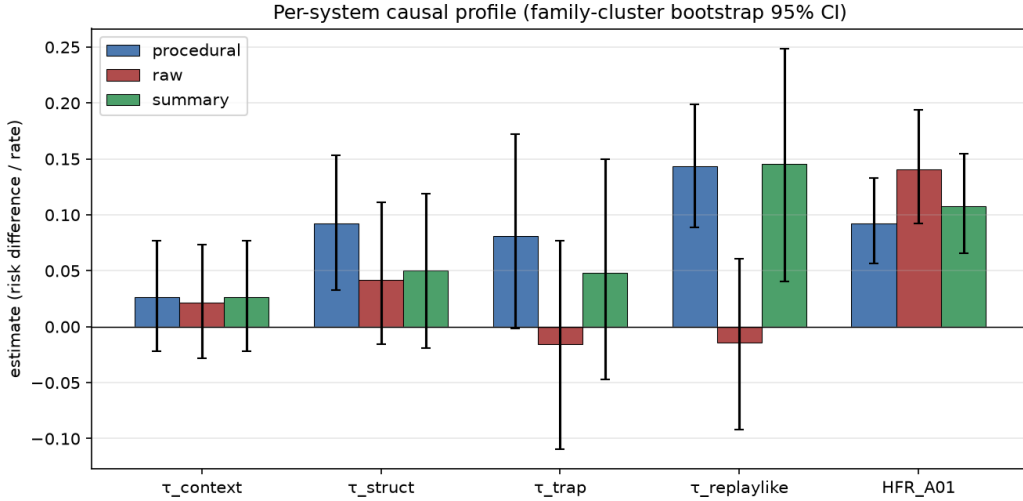


Figure 4: Form/coverage arms on fresh families (H3). Top pad: A10/A11 rates per arm; bottom pad: replay term per arm with 95% CIs. Complete transcripts realize replay materially (TOST rejects $\pm 3\text{pp}$ equivalence), hard-truncated (eco) does not; form contrast is Holm-null in 8 tests.

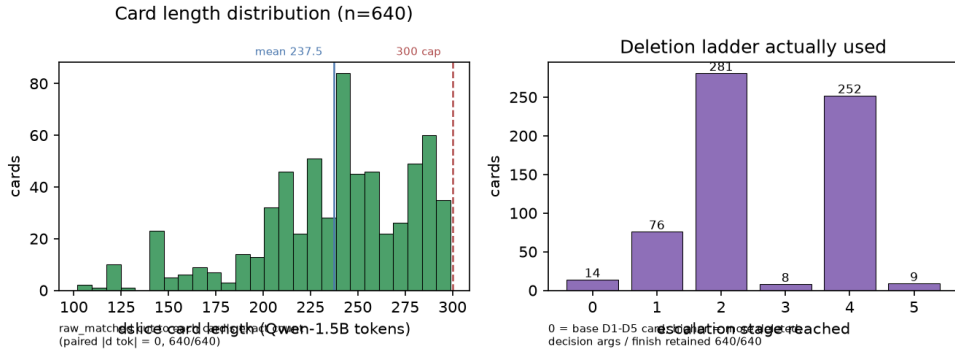


Figure 5: DSLICE card construction audit (Section 7). Left: realized card lengths (mean 237.5 tokens vs. the 300 cap); the paired RAW_MATCHED control is cut to each card’s exact count, so token budgets match per card by construction. Right: how far the deletion ladder had to go to fit budget; decision arguments, aggregate values, and the finish step survive in 640/640 cards (asserted, 0 QA failures).

10.3 LIMITATIONS

- 1. Synthetic environment.** All identification is by construction in one SQLite world with programmatic predicates. Our ALFWorld check (Appendix E) shows the replay/structural ordering repeats at coarse level in a non-synthetic environment but the near-miss harm was contingent on trap construction—directions replicate at aggregate level, magnitudes don’t. We still do not claim AppWorld-scale external validity.
- 2. Narrow model coverage.** Two frozen Qwen2.5 scales drive the main inference; a Llama-3.1-8B pilot replication (same 40 families, 3840 rollouts) confirms replay-dominance (A11 leg = 79% of matched effect), the positive structural signal (7B-like, $\tau_{\text{struct}} = +0.047$ [$+0.020, +0.073$]), and similar flip magnitude (HFR=0.100). Directionally consistent, but the family count remains small, and Llama shows a small positive $\tau_{\text{context}} = +0.034$ absent in Qwen: family idiosyncrasies exist.

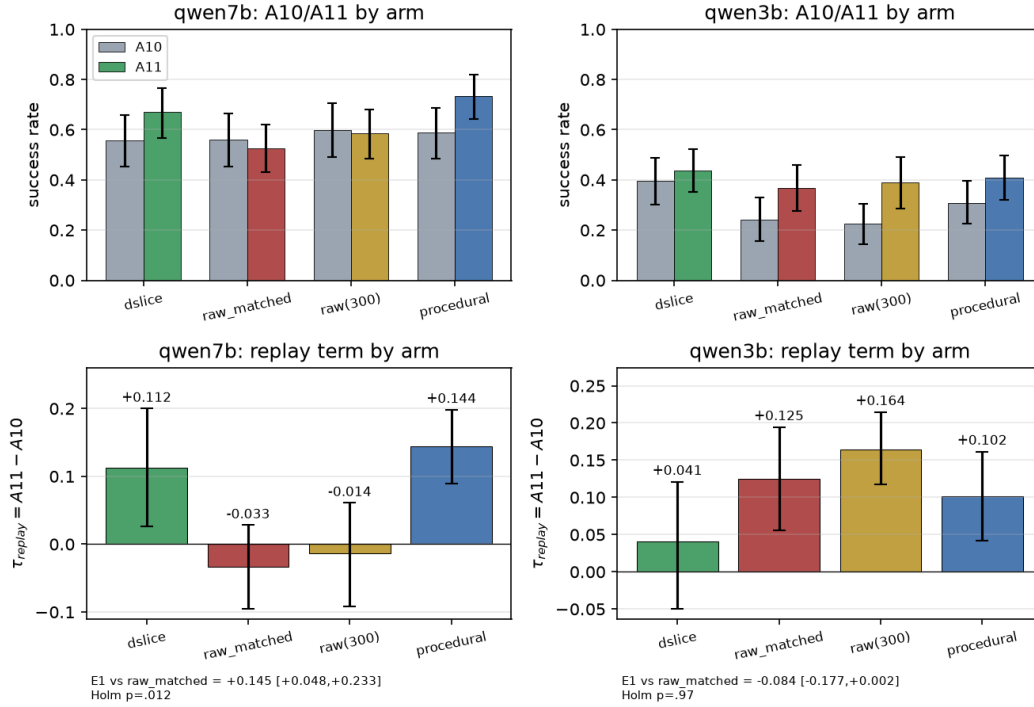


Figure 6: Deployment result (Section 7). Top: A10/A11 rates per arm per model, family-cluster 95% CIs. Bottom: replay term per arm. On 7B, DSLICE realizes replay while exactly-paired truncation realizes none ($E1 = +0.145$, Holm $p = .012$); on 3B the policy is null ($E1 p = .97$)—a capability gate, not a free lunch.

- Equivalence-class ontologies.** Our partial-order classes are stricter than unconstrained "task similarity"; alternative operationalizations (e.g., embedding-based clustering of hidden work-flows) are not evaluated and could reshuffle A10 vs A01 at the boundary.
- Trap-plane surface readability.** Chip-level near-miss detection from text was possible at family hold-out (AUC 0.996); we register it as a semantics of benchmark traps rather than leakage, but it limits what "unreadable label" claims can be made on that axis.
- Analysis breadth.** worry: power is limited by 40/32 family clusters; decompositions at archetype level, though reported, have wide intervals and are exploratory only.

10.4 WHAT SHOULD COME NEXT

The applied sequel is a transportable-memory admission rule validated the same way—we did not run it because it would be governed against standards we cannot yet meet (Section 9). The theory sequel is a distributional-robustness treatment of transformation-shifted memory utility (Angelopoulos et al., 2024; Wu et al., 2016; Jalaldoust & Bareinboim, 2024). And the measurement sequel asks how far these estimates agree with the observational estimators people can only compute without our oracle.

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A H-DC PROTOCOL, PRE-ANALYSIS AMENDMENTS, AND AUDIT

The deployment test of §7 was pre-registered (GATE.PROTOCOL Part IV) and then *amended before any outcome was inspected* (Part IV-A, timestamped 2026-08-08, before the primary grids completed), under the project’s governance rule: implementation gaps were first self-analyzed, then adjudicated with an external reviewer, then registered. Amendments:

- 3B control arm added.** The registration had assumed all controls already existed; the naive-truncation arm in fact existed only for 7B. The 3B RAW arm (2,560 rollouts, identical cards/harness/seeds) was run as a registered correction, keeping E1 estimable at both scales and the Holm family ($m = 6$) intact.
- Paired-budget control made primary.** Realized DSLICE card length (mean 237.5 tokens) is genuinely shorter than the 300-token cap, so a “shorter is conservative” argument would not have sufficed: prompt length itself can affect success. The RAW_MATCHED control—same transcript truncated to the *exact* paired token count (per-card $|\Delta| = 0, 640/640$)—became the primary contrast; RAW(300) is reported as the secondary senior baseline.
- Packaging disclosed.** Action copies are canonical dumps of the parsed action JSON (strips harness parse artifacts; 18/4,510 affected steps, with verbatim fallback wherever a tail could carry recoverable content); exact-duplicate non-mutating steps are deduplicated; the escalation ladder reaches stages $\{0, 1, 2, 3, 4, 5\} = \{14, 76, 281, 8, 252, 9\}$ cards. Claims therefore attach to the DSLICE *package*, not to any single rule.

Execution-environment assertions passed: identical vLLM/torch/GPU versions across arms; config hashes uniform at 7B (3B pilot N/Q rows differ only in a GPU-memory-utilization resource field); the one post-hoc harness change (single-BOS tokenization, introduced for Llama) was verified to be token-id-identical on Qwen2.5, i.e., a no-op for every arm it touches. Inference: aligned family-cluster bootstrap (2000 reps) with the primary Holm family $m = 6$ $\{E1 \times 2$ models, $E2 \times 2$, HFR-noninferiority $\times 2\}$; trap noninferiority uses one-sided bootstrap bounds against the $\pm 5pp$ margins. Full numbers: `pilot/dslice/HDC.RESULTS.json`; protocol text: `GATE.PROTOCOL.md` Part IV/IV-A.

B REPRODUCIBILITY

All artifacts at github.com/anonymous/CausalMemBench upon de-anonymization:

- generator `pilot/generate_families.py` (pilot seed 20260807, H3 seed 20260809); sealed-vs-public split; oracle walker execution logs confirming 800/800 and 640/640 legal terminals;
- environment `pilot/env_relationalops.py`, `DSL_pilot/program_dsl.py`, `harness_pilot/harness.py` (vLLM 0.6.6.post1 / torch 2.5.1+cu124 on RTX A5000 24GB, fp16, temperature 0.7, top-p 0.9, max 12 steps $\times$ 512 tokens);
- rollouts JSONL: 41,216 rows (7,680 pilot + 7,680 H-C + 9,216 H3 + 3,840 Llama-8B replication + 12,800 H-DC deployment grids) with per-row commit, config-hash, model, versions, GPU;
- pre-registered protocols: GATE_PROTOCOL Parts I–IV (and the pre-analysis Part IV-A amendment) dated 2026-08-07/08, all registered before analysis runs;
- audit suite `pilot/audit/{probes,difficulty_robust,continuous_s,equivalence_audit}.py`, blind annotations, judge outputs verbatim; DSLICE/RAW_MATCHED card builders with per-card QA maps;
- analysis: family-cluster bootstrap utilities, Holm/TOST routines, and the exact `analyze.py/json` dumps reported here;
- design registers: RESEARCH_LEDGER (pre-registration chain + external-review outcomes), §13 artifacts, construction rulings for H3 cards.

C FAMILIES AND ARCHETYPES

40 pilot families = 4 archetypes $\times$ 2 domains (crm/inv/ticket/cal) $\times$ 5 instances; 32 H3 families = 4/archetype $\times$ 8 schemas with disjoint parameters. For each family: 4 program-preserving siblings + 1 near-miss + a10 cross-domain partner + a00 unrelated partner, style counterbalanced by a 6-template Latin square, memories 200–300 Qwen-1.5B tokens measured by tokenizer, similarity buckets frozen against pilot distribution anchors.

D SECOND-FAMILY REPLICATION (COMPLETED)

Llama-3.1-8B (NousResearch/Meta-Llama-3.1-8B-Instruct) replicates the identical grid (3840 rollouts, same config/harness). Success rates $N=.373$, $Q=.408$, $A00=.364$, $A01=.452$, $A10=.411$, $A11=.544$. Estimates: $\tau_{\text{struct}} = +0.047$ [$+0.020, +0.073$] (7B-like, opposite 3B); $\tau_{\text{trap}} = +0.087$ [$+0.043, +0.134$]; $\tau_{\text{replay}} = +0.133$ [$+0.069, +0.193$]; $\text{HFR} = 0.100$ [$0.075, 0.125$]; $\tau_{\text{context}} = +0.034$ [$+0.010, +0.058$] (family-specific: Qwen has ≈ 0). Replay-heavy decomposition: the A11 leg carries 79% of the matched effect, mirroring the 7B 72% share. See `pilot/llama8b/LLAMA8B.RESULTS.json`.

E EXTERNAL VALIDITY CHECK (ALFWORLD, COMPLETED)

A coarse effect-ordering check in ALFWorld (Shridhar et al., 2021) (Qwen2.5-7B; 3 task families — clean/, heat/, pick-two — $\times$ 4 siblings $\times$ 4 cells $\times$ 3 seeds, 144 rollouts, max 30 steps, env-verified win) maps pilot cells as $N/R(\approx A11)/S(\approx A10)/X(\approx A01$ with the critical step merely omitted, not actively flipped). Verdict, reported without overclaim: **partially reproduced**. Aggregate ordering $R(0.222) > S(0.194) > N(0.167)$ supports replay-superior-to-structural-superior-to-nothing, at $n=36/\text{cell}$ with overlapping Wilson intervals—direction only, and the clean family alone reproduces the full pilot pattern ($R=S=.33 > X=.08 > N=.00$). **The near-miss harm did not reproduce** ($X=0.250 \geq N$, paired flips 3 harmful vs 6 helpful): our ALFWorld near-miss is a recoverable omission rather than an active polarity/order contradiction, so a 7B model can recover the missing step from the visible goal text. Reading: replay-over-structure ordering survives a non-synthetic env, while harmful near-miss transfer appears contingent on whether the trap actively contradicts the task rather than omits a step. Full construction and deviation log: `pilot/external/EXTERNAL_VALIDATION.md`.